

Eco-friendly erucamide–polydimethylsiloxane coatings for marine anti-biofouling

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ARTICLE INFO

Keywords:

Erucamide
Polydimethylsiloxane
Surface coating
Marine anti-biofouling

ABSTRACT

Marine biofouling of ship hulls and ocean structures causes enormous economic losses due to increased frictional drag. Thus, efforts have been exerted worldwide to eliminate biofouling. In addition, a strong demand exists for the development of a cost-effective and eco-friendly anti-biofouling coating technology. Thus, erucamide-polydimethylsiloxane (EP) coating is proposed in this study. EP exhibits a hydrophobic surface as the erucamide content and drag reduction effect increase. In this study, the drag reduction effect of the EP 2.5 is better than that of glass and polydimethylsiloxane (PDMS) surfaces. Moreover, the proposed EP coatings are observed to prevent the biofouling induced by bacteria (*E. coli*) and brown algae (*Cladosiphon* sp.). In addition, through a marine field test, the anti-biofouling effect of the EP surface is found to be better than the previously studied oleamide-PDMS (OP) surface. In the marine field test, the EP 2.5 demonstrates superior anti-biofouling performance for 5.5 months under real marine environment. The proposed eco-friendly EP coating method could be applicable to marine vehicles that require effective drag reduction and anti-biofouling properties.

1. Introduction

Biofouling is a widespread problem in the maritime industry [1,2]. Biofilms contaminate a wide variety of structural elements, systems, and devices such as pipes, medical implants, food processing facilities, and heat exchangers [1,3]. In marine environments, bacteria attached to a biofilm supply abundant nutrients for further settlement of fouling macroorganisms [4]. The fouling of ship hulls and marine structures increase maintenance expenses and fuel consumption due to high frictional drag and a shortened period of drydocking to remove the fouling organisms over the coating lifetime [5,6]. Therefore, developing new materials is necessary to resolve biofouling problems [2,4,7]. Despite numerous studies over the past decades to develop various anti-biofouling materials for effective anti-biofouling surfaces, prevention of biofilm formation remains a challenge [8]. Conventional anti-biofouling surfaces previously developed were often toxic or the coating substances were easily detached from the coating surface, and their drag reduction effects were not evaluated. A self-polishing coating that contains tributyltin (TBT) has been considered as one of the most

effective coatings to combat marine biofouling. However, TBT is globally banned by the International Maritime Organization (IMO) due to its negative environmental assessment associated with toxicity [9]. Most anti-biofouling strategies are based on the surface peeling, release of biocidal compounds, and the physicochemical modification in the surface design [10]. For example, biocides [11,12], antibiotics [13], and silver nanoparticles and heavy metals [14] were employed to induce anti-biofouling ability. However, whilst very useful, these substances possess a number of fundamental problems, such as bacterial resistance and toxicity [15]. Indeed, most of these substances are banned internationally due to their unacceptable environmental impacts. In addition, self-polishing copolymers [16,17], fouling-release polymers [18,19], liquid-infused surfaces [20,21], organogels [22,23], zwitterionic polymers [24–26], polyethylene glycol [27,28], functional polymer brushes [29,30], functional self-healing materials [31] and multifunctional cellulose [32] have been introduced as coating materials in the fabrication of anti-biofouling surfaces. Although numerous studies over the past decades have attempted to develop effective anti-biofouling materials, the prevention of biofilm formation remains a challenging

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<https://doi.org/10.1016/j.colsurfb.2021.112003>

Received 14 December 2020; Received in revised form 11 June 2021; Accepted 22 July 2021

Available online 26 July 2021

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endeavour [33,34]. Recently, numerous efforts have been exerted to develop eco-friendly or nature inspired anti-biofouling materials as an alternative strategy [4].

Several biopolymers have been assessed as promising drag-reducing and anti-biofouling agents to replace high-molecular weight synthetic polymers to improve safety and ameliorate environmental concerns [35]. The polysaccharides and other high-level polymers secreted by fishes, algae and other marine organisms have drag-reducing features [36,37]. Many water-soluble long-chain polymers, such as mucus or mucilage, exhibit efficient drag reduction capability [38]. For example, the epidermal mucus of several fish species was found to promote drag reduction to get fast swimming speeds [39–41]. The mucus secreted from fish skin can trap water molecules and form a sticky hydrophilic gel [42]. This sticky gel separates fish skins from the attached foulants and functions as a hydrodynamic lubrication layer [43]. In addition, mucus traps or repels pathogens, such as viruses or bacteria [44–46].

The use of natural compounds inspired by natural systems to overcome the toxic problems of conventional coatings has received much attention [47]. As a solution to the limitation of current materials, we propose a biomimetic multifunctional erucamide-polydimethylsiloxane (EP) coating inspired by nature. In this study, we developed a new EP coating to achieve drag reduction and anti-biofouling effects simultaneously. In this study, we have focused on the anti-biofouling properties of fatty acid amides among substances in nature. Primary fatty acid amides have been commercially used as a slip additive for plastic polyethylene films and other polymers to reduce frictional resistance or increase lubricity. These amides are typically found in grass, microalgae, and animals [48]. Fatty acid amides provide slippery surfaces to which biofoulers attach weakly [49]. Benthic organisms on these amide surfaces are readily dislodged by water motion [48].

Erucamide ($C_{22}H_{43}NO$), a type of fatty acid amide, has strong potential as a structural material that constitutes bactericidal surfaces due to its intrinsic antibacterial property [50]. As this compound is isolated from nature, it is readily decomposed in natural environments [48]. This material was first identified in the cerebrospinal fluid of sleep-deprived

and hardened through mixing. In this study, a new EP surface is proposed to employ the advantages of erucamide and polydimethylsiloxane (PDMS). The eco-friendly anti-biofouling properties of this EP surface are experimentally examined.

2. Experimental section

2.1. Materials

Erucamide ($C_{22}H_{43}NO$, MW = 337.59, purity >85.0 %) was obtained from Tokyo Chemical Industry (Japan), PDMS (Sylgard 184) silicone elastomer was purchased from Dow Corning (United States), and toluene ($C_6H_5CH_3$, MW = 92.14, ACS reagent, purity ≥ 99.5 %) was obtained from Sigma-Aldrich (South Korea).

2.2. Fabrication of EP surfaces

PDMS prepolymer (base) and cross-linker (curing agent) were mixed at a 10:1 wt ratio. First, the erucamide solution (in toluene) was prepared in 10 mL volumes at concentrations of 0, 10, 20, 40, and 80 w/v%. Erucamide was dissolved in toluene through sonication at 60 °C for 2 h. Then, 20 g of PDMS solution (mixture of base and curing agent at a weight ratio of 10:1) was pre-loaded in tubes. Each erucamide solution of 10 mL (0, 10, 20, 40, and 80 w/v%) was placed in tubes containing PDMS solution of 20 g. Thereafter, the mixture of PDMS solution and erucamide solution was stirred for 2 h in a vortexing mixer. The prepared mixtures were poured on the various substrates such as a glass slide, a cover glass, and an aluminum plate. The test samples were solidified in an oven at 60 °C for 24 h by evaporating toluene and hardening the PDMS-erucamide mixture. Finally, the EP surfaces with the erucamide content of 0.625, 1.25, 2.5, and 5 wt% were prepared, and they were denoted as EP 0.625, EP 1.25, EP 2.5, and EP 5, respectively. The content of erucamide is defined as follows:

$$\text{Weight content (wt\%)} = \frac{\text{weight of erucamide}}{\text{weight of PDMS mixture solution} + \text{weight of erucamide}} \times 100$$

cats. It has also been detected in the cerebrospinal fluids of rats and humans. Erucamide has been known to stimulate the growth of blood vessels [51] and regulate fluid imbalance in animal organs [48]. Erucamide has a high melting point (77.50 °C) and good stability at a high temperature [48]. In addition, oleamide, a type of fatty acid amide, can be melted in a small amount of water. However, erucamide cannot be dissolved in water [52].

Polydimethylsiloxane (PDMS)-based material is one of the promising candidates for nontoxic anti-biofouling applications [53,54]. PDMS is a silicone material with low surface energy and small young's modulus [55–57]. This material can reduce adhesion force that acts between fouling organisms and the coating surface, thereby resulting in easy removal of the fouling organisms [54,58,59]. In addition, the surface can be deformed by dynamic movement of surrounding water, causing the attached biofoulers to be easily shed under turbulent flow conditions [60].

Erucamide has amide functionality beside long alkyl chain, whose functional group has limited activity for organic substitution reaction. There is no chemical bond between erucamide and PDMS. However, the two chemicals are well mixed together and become solid through hardening process. Erucamide and PDMS both have hydrophobic properties. In addition, erucamide does not alter the amorphous structure of PDMS [61,62]. Therefore, these materials can be easily combined

PDMS and erucamide were not chemically combined. They were physically mixed and solidified. This was confirmed by comparing 1H NMR spectra of erucamide, EP 5, 2.5, 1.25, 0.625, and PDMS. The related NMR results are included in Fig S1 of supporting information. The prepared PDMS and EP mixture solution of 380 ± 40 mg was applied to the substrate surface of 1 cm^2 in physical dimension by using a disposable pipette, and the coating thickness was approximately $450 \pm 50 \text{ }\mu\text{m}$.

2.3. Surface morphology and roughness analysis

The morphological features of the PDMS and EP surfaces were examined through atomic force microscopy (AFM, Veeco Dimension 3100). Then, 3D reconstruction of the captured AFM images and subsequent quantitative surface roughness analysis were conducted using Nanoscope V software (version 7.0).

2.4. Surface characterization

The EP surfaces were prepared by mixing at different weight ratios (160:1, 80:1, 40:1, and 20:1) of PDMS (g):erucamide (g) to investigate the effect of mixing ratio on the physicochemical properties of the EP-coated surfaces. The wettability of the fabricated EP surfaces was

evaluated by measuring their water contact angle (CA). The static CA of a 3 μ L deionized (DI) water droplet was measured by using a video-based optical imaging system (Femtofab SDL-TEZD, Korea). The CA of each sample surface was measured independently at five random points. The measured CA values were statistically averaged [55].

The tilting hysteresis of the fabricated EP surfaces was estimated as the surface with a drop of DI water is automatically tilted. The advancing CA (θ_{adv}) and receding CA (θ_{rec}) of each sample surface were measured at three random points. The measured CA values were statistically averaged.

The pressure drops in microchannel flows over the fabricated EP surfaces were measured as a function of Reynolds number to examine the drag reduction performance of the EP surfaces. The drag reduction of the proposed EP surfaces was examined by direct measurement of pressure drops in channel flows. Pressure drops in a rectangular PDMS microfluidic channel were measured to estimate the reduction of the frictional drag exerted on the fabricated EP surfaces. A glass slide coated with the PDMS or EP was attached to the microchannel (length: 70 mm, width: 4 mm, and height: 0.6 mm). The inlet of the microchannel was connected to a syringe pump (PHD ULTRA, Harvard Apparatus, USA). Its outlet was attached to a waste reservoir by a teflon tube. Two pressure taps were installed at the microchannel and connected to a differential manometer (PX409-10WDWUUSBH, Omega Engineering, USA). Differential pressure of the two taps was recorded when the channel flow reached a steady-state condition. The pressure drops over the EP surfaces were measured with varying flow speeds to assess the frictional drag reduction of the surfaces. The drag forces exerted on the PDMS and glass surfaces were also measured as control references for comparison with the drag reduction effect of the EP surfaces [55,63].

2.5. ATR-FTIR analysis

The chemical composition of the EP surfaces was analyzed using an attenuated total reflection Fourier-transform infrared (ATR-FTIR) spectroscopy. ATR-FTIR experiments were performed at the 12D IRS beamline of Pohang Accelerator Laboratory (Pohang, Korea). The FTIR spectra were obtained by using a VERTEX 80 V spectrometer (Bruker, USA) attached with an ATR accessory (PIKE Technologies, USA). The spectra were recorded at a resolution of 4 cm^{-1} by using a deuterated triglycine sulfate detector under a vacuum condition [55,64].

2.6. Antibacterial assay

Escherichia coli (*E. coli*), a gram-negative bacteria, was used as a model to evaluate the antibacterial property of the EP surfaces. *E. coli* (ATCC25404) was purchased from the American Type Culture Collection (Manassas, VA). The bioassay procedure began with a stock solution of *E. coli* streaked onto tryptone and yeast extract-dehydrated agar plates. The plates were placed in an incubator at 37 °C overnight and stored at 4 °C until use. A single colony was picked from streak plates and inoculated into 5 mL sterile Luria broth (LB Broth Miller, BD Difco, USA) to prepare *E. coli* broth. Then, *E. coli* was grown for 24 h by shaking at 200 rpm at 37 °C. After incubation, the *E. coli* solution was diluted to 1/5 into the sterile LB broth. Glass cover slips (12 mm ϕ) coated with PDMS and EP were placed in 12-well plates. Then, the samples were incubated with 2 mL bacterial solution for 18 h at 37 °C. After incubation, the samples were rinsed with phosphate-buffered saline (PBS) buffer to remove the nonadherent bacteria. The test samples were stained by using a live/dead bacterial viability assay kit (BacLight L7012, Molecular Probes, Invitrogen, USA). The stained samples were rinsed with PBS buffer. The fluorescence images of stained samples were captured by a Leica confocal microscope (TCS SP5II MP; Leica Microsystems, Germany) with a 40 $\times$ objective lens (Leica). The acquired images were analyzed using LAS AF 2.7 software (Leica). Bacteria were quantified by cell counting (fields of view = 123.02 μm x 123.02 μm). The number was calculated from three replicates, and the results were

expressed as mean $\pm$ standard deviation.

To capture scanning electron microscope (SEM) images, we fixed the adherent bacteria by 10 % formalin solution. The fixed samples were rinsed with PBS buffer and DI water. Subsequently, the dried samples were coated with platinum (SC7640, Quorum Technology, United Kingdom) and examined through SEM (JEOL JSM-7401 F, Japan) at an acceleration voltage of 15 kV. The multistep procedure of the anti-biofouling experiment is illustrated in Fig. S2a.

2.7. Brown algal anti-biofouling assay

The biofouling properties of the submerged glass, polystyrene, and proposed EP surfaces containing various amounts of erucamide were examined by exposing them to brown algae (*Cladosiphon* sp.) and capturing optical images of anti-biofouling phenomena on their surfaces. An adhesion test on brown algae was conducted to evaluate the anti-biofouling properties of the fabricated EP surfaces. The multistep anti-biofouling experiment process is illustrated in Fig. S3a. *Cladosiphon* sp. was incubated in IMR medium at 20 °C under an illumination condition of 20 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$ (12 h light:12 h dark). In the anti-fouling experiment, we used a haploid gametophyte of *Cladosiphon* sp. (Fig. S3b). The gametes were finely chopped, and the upper solution was removed through centrifugation. The chopped brown algae were then washed thrice. Then, the test sample was diluted to 1 mg/mL with the IMR medium. Glass cover slips (12 mm ϕ) coated with PDMS and EP mixtures were placed on a 12-well plate. Biofouling characteristics of brown algae on the glass and polystyrene surfaces were observed on the glass cover slip (12 mm ϕ) and empty surface (polystyrene surface) of the plate. Then, 5 mL of the diluted solution was divided into another 12-well plate. Thereafter, the test sample was cultured for 48 h at 20 °C under 20 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$. The medium was replaced to wash the nonadhesive ones at 20 °C under 20 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$. The incubation and observation procedures were continuously conducted for three weeks. The brown algae were observed using a stereomicroscope (SZX10, Olympus, Japan) attached to an IP camera (8000, Hanwha Techwin, Korea). ImageJ software (NIH, USA) was used to analyze the attached algae. The canopy cover was evaluated using ImageJ software. It was calculated from three replicates, and the results were expressed as mean $\pm$ standard deviation.

2.8. Marine field test

The EP 2.5 (2.5 wt%), OP 5 (5 wt%) and PDMS were coated on flat acrylic plates of 100 mm x 100 mm x 3 mm in size under 26 °C of temperature and 44 % of relative humidity. A clean acrylic plate without any coating and an acrylic plate coated with pristine PDMS were used as the reference and control surfaces, respectively (Fig. S3). Marine field test was conducted for 5.5 months (from 6 March to 19 August in 2020) in the West Sea (coordinates 36°08'13.6"N 126°32'19.3"E) in front of Seochen town, Chungnam province, Korea. The test places were located next to a kelp farm, where marine organisms and seaweeds are easily attached. The test samples were installed at a frame (Fig. S3a). The surfaces coated on the substrates were directly exposed to seawater. The prepared test samples were immersed in seawater at a depth of 1 m from sea level. A hydrostatic pressure of 1.16 bar was applied to the test samples during the field test. The salinity and temperature of the surrounding seawater were ranged from 29.5–32.1 psu (practical salinity unit) and from 5.6–21.0 °C during the test, respectively. The pH level of seawater was 8.1 $\pm$ 0.1. Deposition of marine organisms and seaweeds on the test samples were photographed with a digital camera (Fig. 1).

2.9. Statistical analysis

Statistical analysis was conducted using GraphPad Prism 8 software (GraphPad, USA). Shapiro–Wilk test was employed to determine the data distribution. Statistical significance was determined by one-way

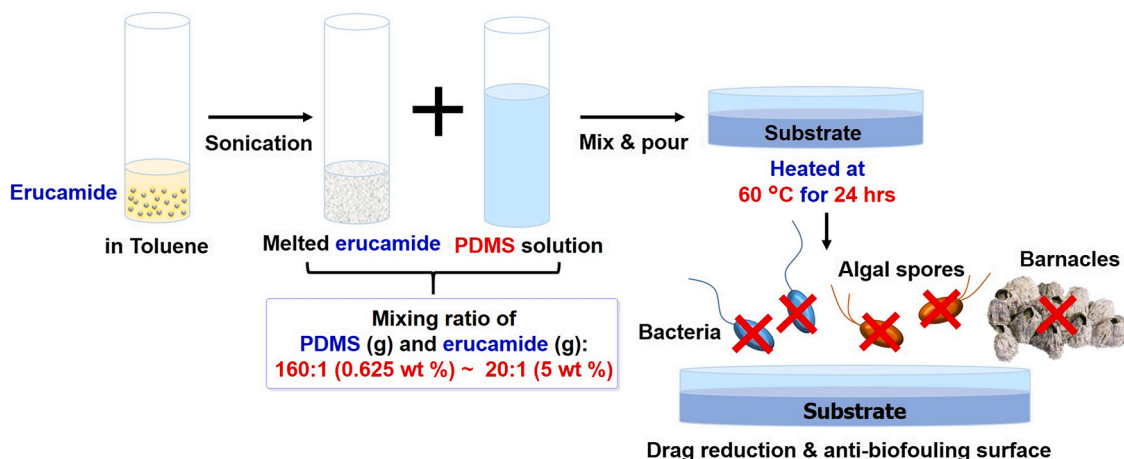


Fig. 1. Illustration of fabrication procedure of an erucamide-PDMS (EP) surface by mixing erucamide and PDMS solution.

Welch's ANOVA with Dunnett's T3 *post hoc* test. In some cases, the two-tailed unpaired Student's *t*-test (with Welch's correction) was adopted to compare two groups. Statistical significance was accepted when $P < 0.05$.

3. Results and discussion

3.1. Surface morphology and roughness

Fig. 2a shows typical 3D AFM images of the PDMS and EP surfaces with a scan size of $50 \times 50 \mu\text{m}$ for each sample. The captured AFM images show that as the portion of erucamide contained in the EP increases, so does the surface roughness. Variations in the surface roughness of the tested coating surfaces were expressed by the following root-mean-square (RMS) values:

$$\mathcal{R}_{rms} = \sqrt{\frac{1}{n} \sum M^2}$$

where n is the number of samples, and M is the measured thickness of coating layer [65].

As shown in Fig. 2b, the higher the content of erucamide in the EP, the larger the RMS values of the surface. The volume of erucamide increases as the amount of erucamide-PDMS mixture increases, and the crystalline size of erucamide in the mixture becomes non-uniform. Thus, for EP 5 surface (5 wt %, PDMS:erucamide = 20:1), the erucamide is heterogeneously distributed in PDMS elastomers. EP 5 contains more erucamide in the same amount of solvents (toluene), compared to other EPs 2.5 ~ 0.625. In addition, the amount of erucamide mixed in the

PDMS solution is higher in EP 5 than that of other EPs 2.5 ~ 0.625. Thus, the crystalline size of EP5 is larger than other EPs 2.5 ~ 0.625, and erucamide may be coalesced into them. Based on these properties, the EP 5 surface is estimated to be more rough compared to other EP samples. These surface morphological features of the fabricated EP surfaces may provide various surface functionalities such as water contact angle, anti-biofouling, and drag reduction effects.

3.2. Surface characterization

Wettability is one of the important factors that affect the anti-biofouling properties of material surfaces. Initially, we compared the static water contact angles (CAs) of the PDMS and EP surfaces (Fig. 3a). The water CA increases as the erucamide content in the mixture increases. However, the EP 5 surface is less hydrophobic than the EP 2.5 surface. Although the EP 5 surface contains more erucamide than the EP 2.5 surface, the contact angle is smaller than the EP 2.5 surface. This result may be attributed to the significantly higher roughness of the EP 5 surface compared with other EP surfaces tested in this study because the erucamide particles in the EP 5 sample are uneven. As shown in Fig. 2b, the RMS value of the EP 5 surface is three times larger than the EP 2.5 surface. The surface heterogeneity is proportional to the RMS value of the surface roughness.

The advancing CA (θ_{adv}) values do not change largely in all the samples when the bottom is tilted at approximately 20° (Fig. 3b). However, as the bottom tilts more than 20° , the θ_{adv} for all the samples is gradually decreased as the tilt angle increases. The CA reduction is due to the slippage of water droplets on the EP surface. The water droplets

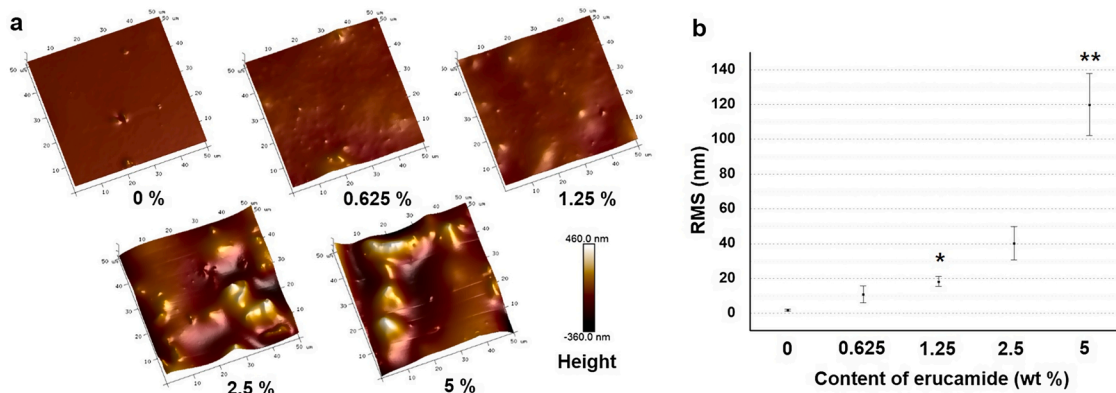


Fig. 2. Comparison of surface morphologies of the PDMS and EP surfaces. (a) 3D AFM images of the fabricated PDMS and EP surfaces. (b) Comparison of root-mean-square (RMS) values of surface roughness. Data are expressed as mean $\pm$ SD (one-way ANOVA vs. PDMS, * $P < 0.05$, ** $P < 0.01$, $n = 5$).

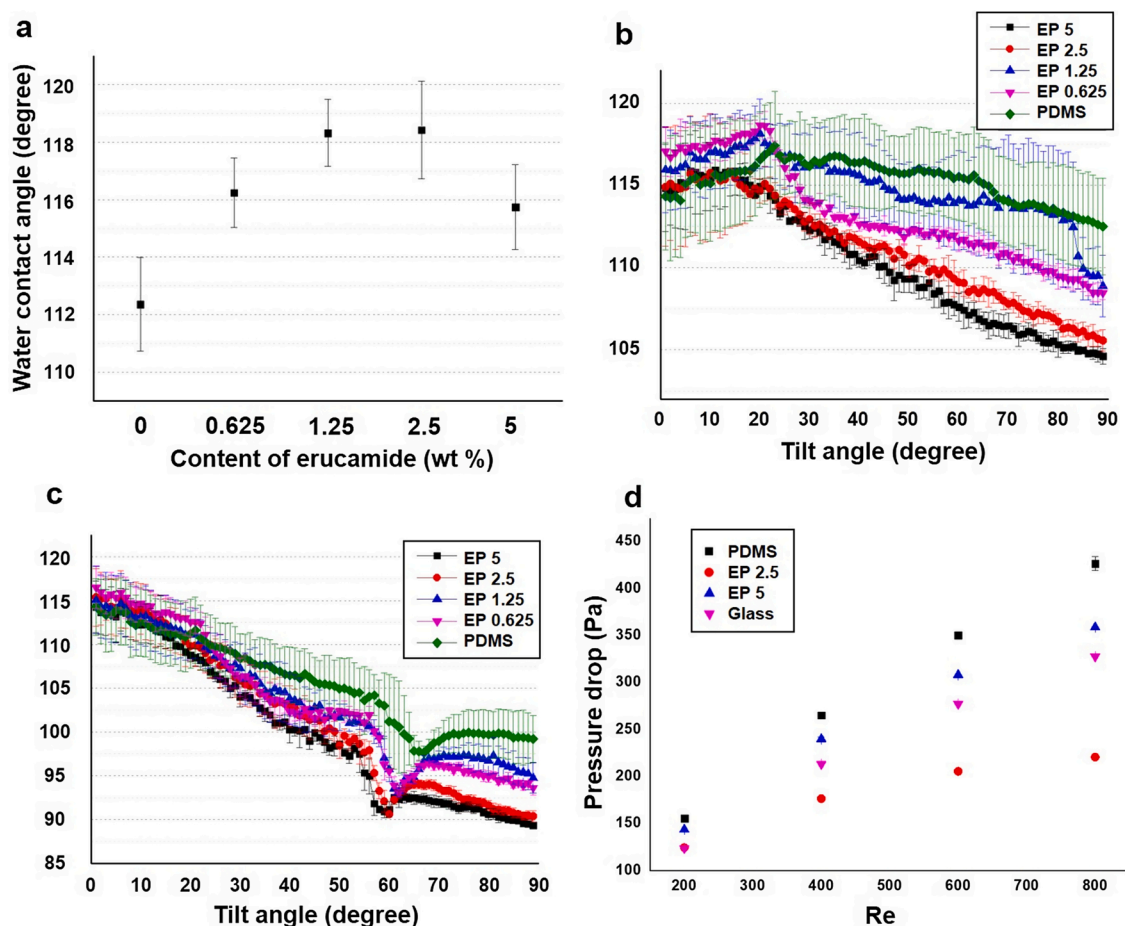


Fig. 3. (a). Comparison of water contact angles on the PDMS and fabricated EP surfaces ($n = 5$). Tilting hysteresis of (b) advancing contact angles (θ_{adv}) and (c) receding contact angles (θ_{rec}) of the PDMS and fabricated EP surfaces ($n = 3$). Error bars represent the standard deviation. (d) Effect of the content of erucamide in EP on pressure drop of the EP surfaces attached to microchannel wall according to Reynolds number ($n = 3$). Working fluid is DI water.

slip better on the EP 5 and EP 2.5 surfaces than the other samples. As illustrated in Fig. 3c, a similar tendency is shown in the receding CA (θ_{rec}). Finally, for the EP 5 and EP 2.5 surfaces, the water droplets slip more easily on the surface as the tilt angle increases. These results imply that depending on the content of erucamide in EP, the degree of interaction between the EP surface and water may be different.

Underwater drag reduction plays an important role in mitigating marine biofouling. The drag forces exerted on the pristine PDMS and glass surfaces were also measured for comparison with the drag reduction effect of the EP surfaces. The glass surface was chemically stable and its drag reduction effect was better than that of the PDMS surface. The measured pressure drops for the EP 2.5 and EP 5 surfaces were smaller than those of pristine PDMS at all Reynolds number conditions tested in this study (Fig. 3d).

The surface roughness increased as the content of erucamide mixed with PDMS increased. However, drag did not decrease in proportion to the erucamide content. The cases of EP 2.5 and EP 5 showed less friction drag than PDMS. In particular, the friction drag of EP 2.5 was less than that of glass. This result would suggest that the effects of surface roughness and the ratio of erucamide content on the friction drag reduction were interconnected in a complex manner. The best drag reduction of EP 2.5 implied that the mixing ratio was ideal to provide a superior drag-reduction effect. This result would indicate that the erucamide could induce less frictional drag due to interfacial slippage. As the Reynolds number increased, the drag reduction effect of the EP 2.5 became smaller than that of the glass surface. The greater drag-reduction effect of the EP 2.5 than that of EP 5 was attributed to its significantly less roughness, as shown in Fig. 2b.

As the EP had a slippery surface in water, organisms adhered on the EP surface could be easily detached even at a slow flow condition. This slippery property may act as the main mechanism that causes effective anti-biofouling ability of the proposed EP surface, which prevents the attachment of fouling organisms in addition to the drag reduction effect.

In addition, the incorporation of erucamide content into the PDMS carrier phase varies, according to erucamide amount (CA measurements). This homogeneity problem should be solved before curing step to enhance surface features as well as anti-biofouling efficiency. Through a further study, we will solve this problem in the near future by finding pertinent solvents and ways to better dissolution without forming erucamide crystals.

3.3. FTIR spectra

The presence of erucamide on the fabricated EP surfaces was successfully mapped through functional group analysis using synchrotron-based ATR-FTIR spectroscopy. The IR band positions and attributions of PDMS were used as the base in the interpretation of functional coating. The spectra were divided into two groups of similar functional characteristics. The Si—O—Si stretching bands at 1020 cm^{-1} and 1074 cm^{-1} indicate the presence of PDMS in the fabricated EP surfaces (Fig. 4a) [66]. The peaks of double bonds such as C=C and CO= indicate the presence of erucamide on the EP surfaces (Fig. 4b) [67]. The EP surfaces contain hydrophobic components such as CH, CH₂, and CH₃ (Fig. 4c) [68]. The peak of N—H also indicates the presence of erucamide on the EP surfaces (Fig. 4b and d) [69].

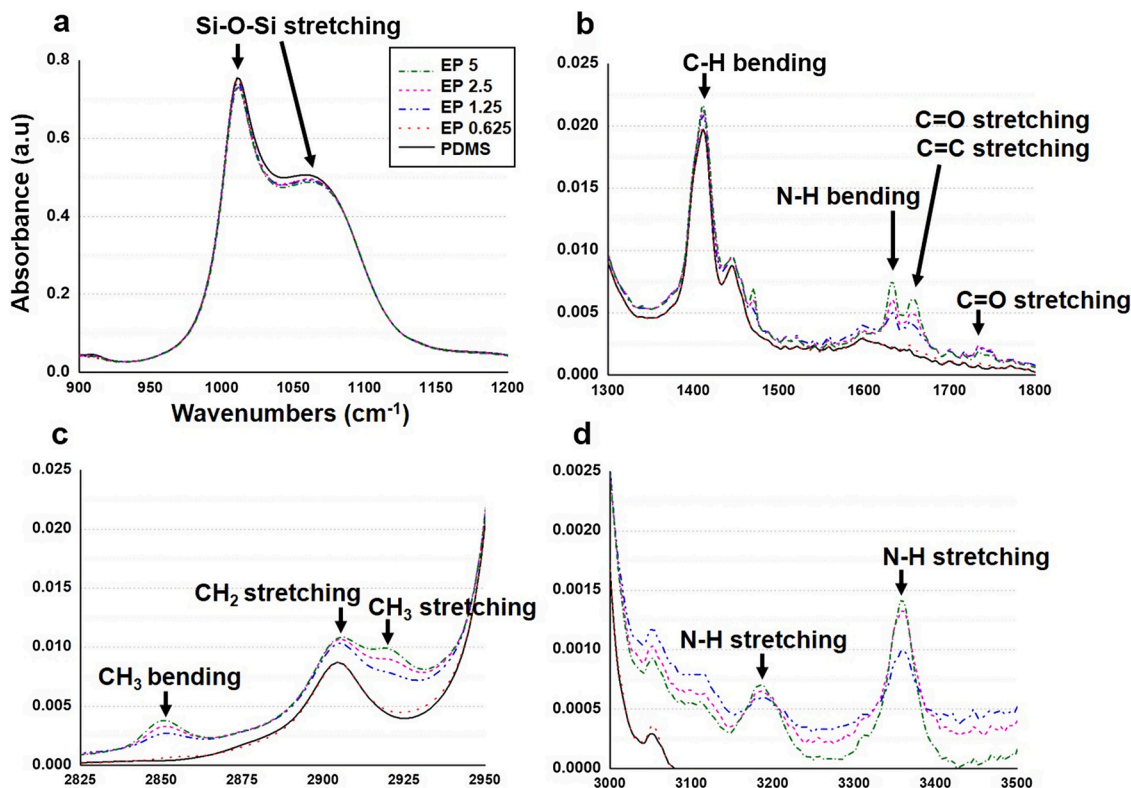


Fig. 4. (a–d). FTIR spectra of the PDMS and EP surfaces at four different wave number ranges.

3.4. Antibacterial properties

The properties of *E. coli* are similar to those of most marine bacteria and have been extensively studied [70–73]. Therefore, *E. coli* was used as a model organism in this study. Fig. S2c shows fluorescence images of *E. coli* cultured on the PDMS and EP surfaces. Living cells display green fluorescence and dead cells show red fluorescence. The total number of bacteria on the PDMS surface is much larger than that on the EP surfaces. Compared with a large number of bacteria on the pristine PDMS, much fewer bacteria are observed on all the EP coating surfaces tested in this study (EP 0.624 ~ EP 5) (Figs. 5, S2b and S2c). As a future study, we are planning a detailed study on the anti-biofouling effects of EP surfaces, especially the antibacterial properties of erucamide itself. For this,

we will examine the anti-biofouling effects of erucamide for a wide variety of bacterial species not covered in this study.

3.5. Anti-biofouling properties of brown algae

As illustrated in Fig. 6a, a large number of brown algae (filamentous cells, haploid gametophytes of *Cladosiphon* sp.) adhere to the control (glass, polystyrene, and PDMS) surfaces. As shown in Fig. 6b and c, the anti-biofouling effect of the EP surfaces is enhanced as the amount of erucamide contained in the EP samples increases. In particular, the EP 5 surface exhibits excellent anti-biofouling performance.

3.6. Marine field test

To test the long-term durability of anti-biofouling performance of the proposed EP coating under a natural marine environment, a marine field test was conducted. Among the samples tested in this study, EP 2.5 was found to have the most hydrophobic surface and exhibit the best drag reduction effect. In addition, it shows excellent anti-biofouling property. Thus, the EP 2.5 was chosen as the test specimen in the marine field test. To compare the drag reduction and anti-biofouling effects of EP with those of oleamide-PDMS (OP), we also tested OP5 surface that has been estimated to be most effective among the previously evaluated OP surfaces.

Fig. 7 shows typical images of the tested panels after immersion in natural seawater for 3 and 5.5 months. As a result, the acrylic plate (3 months) without any coatings is observed to be fully covered with slime and marine organisms, including microorganisms and microalgae of the testing site. The PDMS panel is also covered with slime. However, almost no fouling organisms are observed on the panels coated with EP 2.5 and OP 5. Especially, the biofouling performance of EP 2.5 surface is relatively better than OP 5, acrylic and PDMS surfaces. After 5.5 months, the tested EP and PC surfaces are relatively clean compared to acrylic and PDMS surfaces. This confirms the long-term anti-biofouling property of erucamide and oleamide. Especially, EP 2.5 demonstrates outstanding

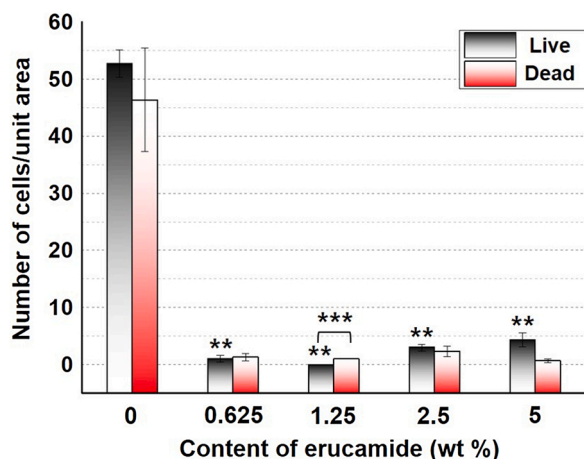


Fig. 5. Comparison of numbers of live and dead cells attached to the PDMS and EP surfaces. The unit area is $123.02 \mu\text{m} \times 123.02 \mu\text{m}$. Data are expressed as mean $\pm$ SD (one-way ANOVA vs. PDMS and *t*-test, $^{**}P < 0.01$, $^{***}P < 0.001$, $n = 3$).

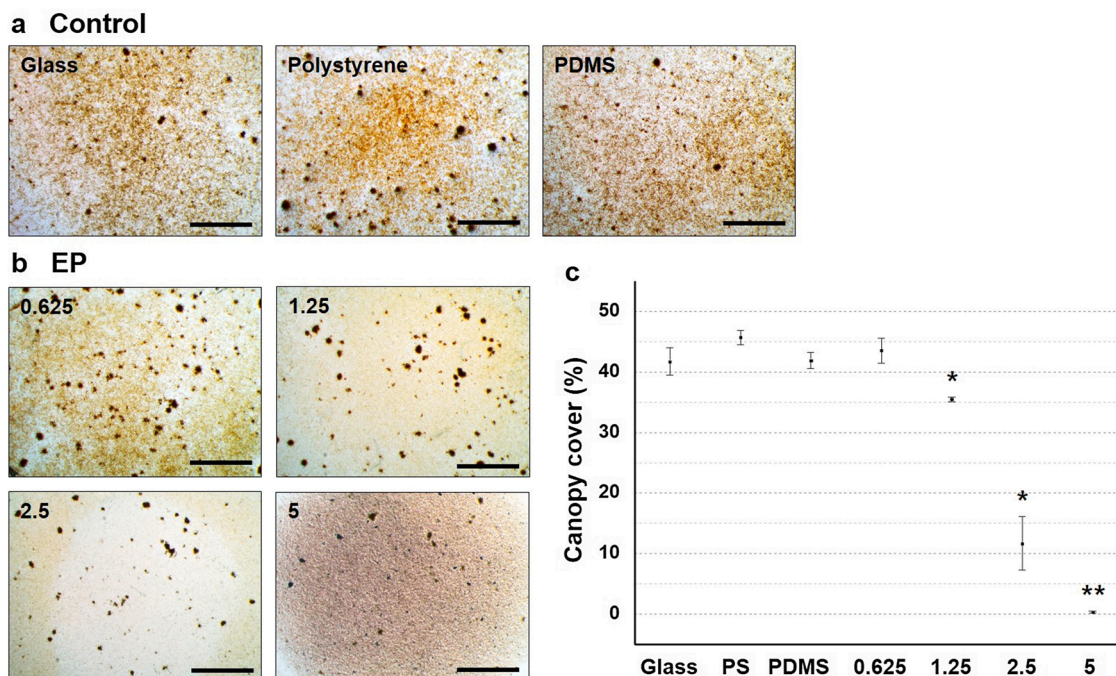


Fig. 6. (a, b). Typical images of brown algae (*Cladosiphon* sp.) cultured on the control (glass, polystyrene, and PDMS) surfaces (a) and fabricated EP surfaces (b) for 3 weeks. (c) Comparison of number of brown algae cultured on glass, polystyrene, PDMS, and EP surfaces for 3 weeks. Scale bars = 2 mm. Data are expressed as mean $\pm$ SD (one-way ANOVA vs. PDMS, * $P < 0.05$, ** $P < 0.01$, $n = 3$).

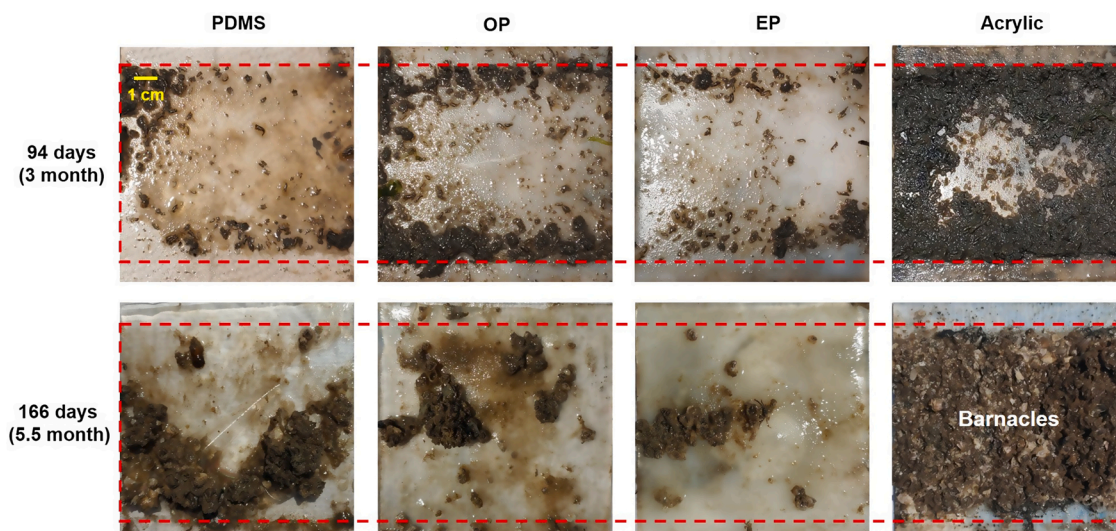


Fig. 7. Photos of the tested panels coated with PDMS, OP 5 and EP 2.5 and acrylic plate after immersion in natural seawater for 3 and 5.5 months. The border region (outside of red dotted box) of the panels was excluded in the analysis, because the region was made of acrylic part for panel insertion.

sustainability of high anti-biofouling property.

3.7. Surface energy of EP surfaces for anti-biofouling

The surface energy of EP surfaces was estimated by using the following equation based on the measured contact angles (CA) and surface tension of water [74]:

$$\cos \theta = -1 + 2\sqrt{\frac{\gamma_s}{\gamma_w}}[1 - \beta(\gamma_w - \gamma_s)2]$$

where θ denotes the water contact angle, γ_s is the surface energy of the substrate, γ_w is the surface energy of water (72.8 mJ/m^2), β is a constant value of $1.057 \times 10^{-4} \text{ m}^2/\text{mJ}$. The above equation indicates that a

substrate with a high CA has low surface energy. The low surface energy is closely related to low adhesion of marine biofoulants [75]. The PDMS and EP surfaces have low surface energy due to their hydrophobicity, compared to the hydrophilic acrylic surface. The surface energy of EP surfaces was ranged from $14.3\text{--}14.6 \text{ mJ/m}^2$, as illustrated in Fig S5. PDMS has a surface energy of 19.8 mJ/m^2 . According to the Baier curve (bioadhesion curve versus surface free energy in the range of $10\text{--}70 \text{ mJ/m}^2$), the minimum bioadhesion does not occur at the lowest surface energy, but occurs in the optimum surface energy range (from $20\text{ to }30 \text{ mJ/m}^2$) [76]. The surface energy values of the proposed EP surfaces belong to the optimum surface energy range for minimum bio-attachment. Accordingly, in terms of surface energy, the EP surfaces are more suitable for achieving anti-biofouling property, compared to OP or

PDMS surfaces.

4. Conclusions

The proposed EP surfaces exhibit drag reduction effect. EP 5 and EP 2.5 surfaces are more effective in reducing drag compared with PDMS surface. In particular, the drag reduction of EP 2.5 is better than that of glass surface. Furthermore, compared with the pristine PDMS coating, the proposed EP surfaces effectively prevent the adhesion of bacteria (*E. coli*) and brown algae (*Cladosiphon* sp.). In addition, a marine field test demonstrates that EP 2.5 coating has superior anti-biofouling performance for 5.5 months under real marine environment. The field test result of EP 2.5 is better than that of OP 5. These results show that the proposed EP surfaces have better anti-biofouling effects than OP surfaces. Thus, they can be utilized as an effective anti-biofouling coating in marine environments. Conclusively, the proposed eco-friendly EP surface is expected to be applicable to marine vehicles and industrial operations that require drag reduction and anti-biofouling features.

CRedit authorship contribution statement

Eunseok Seo, Ji Woong Lee, Dohoon Lee, Myeong Ryun Seong: performed experiments, analyzed data. **Eunseok Seo, Ji Woong Lee, Dohoon Lee:** wrote the manuscript. **Gwang Hoon Kim:** provided guidance and technical expertise on anti-biofouling experiment and marine field test. **Dong Soo Hwang:** provided guidance and technical expertise on erucamide. **S.J.L.** supervised the project and finalized the manuscript. All authors contributed to reading and editing the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors are grateful to Kyunghum Baek and Chang-yeol Oh for their valuable assistance in the imaging experiment conducted at the SEM and AFM laboratory of the National Institute for Nanomaterials Technology. The authors also thank the staff and supporting group of the 12D X-ray beamline of the PAL (Pohang, Korea) for their assistance in the FTIR experiment. This study was supported by the National Research Foundation of Korea through a grant funded by the Korean Government (MSIP, No. 2019M3C1B7025088).

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.colsurfb.2021.112003>.

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